

SLIDE 1 — title (camera on)

Hi, I'm Jacob Brams, student number s224927, presenting my simulation study of the FermaCore tablet line for course 42417. The line runs fifty batches of three products, and my question is simple: where is the bottleneck, and can the line meet 2030 demand without new capacity? Let me show you the model.

SWITCH TO ANYLOGIC — flowchart

Each agent here is one 225-kilo batch that becomes about 300,000 tablets, and fifty of them run on the plan. Batches arrive on the left and are routed to one of two blenders for premix. A failure check scraps a few of them. The rest are granulated and then dried — and drying is the slowest step, sharing a single cleaning team with the whole line. After that comes compression on three presses, where ten percent of tablets are lost as scrap, then optional coating, and finally packaging before the bottles ship out.

SWITCH TO DASHBOARD — model running

Running it live, these bars are the real-time utilisations. The dryer climbs to the top and stays there while everything else sits well below — which matches the capacity calculation that predicted drying as the bottleneck, so the model is validated. And down here, work-in-process builds up early and drains out by the end of the run.

SLIDE 2 — KPIs / baseline

Let me explain the five KPIs I chose and why. My main one is throughput — the number of bottles shipped per run — because that is the output the business ultimately cares about; at baseline it's about sixty thousand five hundred.

The second is dryer utilisation, my bottleneck indicator: it shows which station is the binding constraint, and at ninety-four percent the dryer is clearly it. The third is mean flow time — how long a batch spends in the whole line — which captures congestion and lead time, around a hundred and seventy-nine hours here.

The fourth is on-time delivery, my measure of schedule adherence — whether batches meet their release dates. And the fifth, a secondary one, is scrap rate, which tracks material efficiency, or yield.

Together they cover output, the constraint, congestion, service, and waste — and every one is measured directly from the model each run, so all my scenarios are compared on exactly the same footing. This baseline is the reference everything else moves against.

SLIDE — experiments (DOE + bottleneck shift)

My first key result is that no capacity lever moves throughput. I ran a full factorial on adding a dryer, cleaning, and packaging, and none of them is significant. Output is fixed by batches times yield, not by station capacity. And if you do add a second dryer, the bottleneck doesn't disappear — it just moves to the shared cleaning team, which jumps to ninety-five percent.

SLIDE — 2030 demand

My second result is about 2030. As demand grows, the line does produce the higher volume — bottles roughly double — but service collapses. Mean flow time goes from a hundred and seventy-nine hours up to three hundred and thirty-eight, and

on-time delivery falls from a hundred percent down to fifty-four.
The single dryer saturates.

SLIDE — Industry 4.0 + recommendation

So the only lever that actually lifts output is Industry 4.0.
Predictive maintenance at the press cuts scrap from ten percent down to four — that's about plus eight percent, roughly four thousand eight hundred more bottles per run, with no new line capacity at all.

My recommendation is clear: skip the capital expansion, invest in the press sensors now, and decouple the dryer cleaning from the shared team before 2030. Thank you for watching.